

Chapter 60 — Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

- Surgery for **disorders of sex development (DSDs)**, genital atypia and cloacal anomalies is among the **most complex problems in paediatric urology** and among the most debated topics in medicine. Most children with a DSD have some genital atypia.
- **Terminology is itself contested:** most patients with congenital adrenal hyperplasia (CAH) and their parents dislike both "DSD" and "intersex".
- The birth of a child with a DSD is highly stressful. The team must explain the **diagnosis and every controversy and option**.
- **The historical default was early "normalising" surgery.** Parents found "doing nothing" stressful and viewed surgery as obvious rather than a decision. Against this, **genital proportions vary widely in normal children and adults**. Families should also hear about the child's possible long-term psychosocial experience.
- **Non-negotiable principles:** present all pros and cons of surgery and of no surgery; **no parent or child is ever talked into surgery**; the team supports the family whichever path is chosen; families have access to support groups (CARES Foundation, Accord Alliance, Magic Foundation, InterACT).
- The controversy is **not the technique but the timing, and who decides** (parents or child). The authors are neither for nor against surgery; they advocate a **multidisciplinary family-centred team**. The chapter describes the surgical options at any age and assumes family and team agree on the plan.
- **Minimum team:** patient and family, paediatric urologist or surgeon, paediatric endocrinologist, geneticist, neonatologist, psychologist or psychiatrist, and ethicist. Ideally also patients who have and have not elected surgery, access to all advocacy groups, and clergy if desired.

Concept — *The dispute in DSD surgery concerns **consent and timing, not operative technique**. The multidisciplinary team, unbiased counselling and equal support for either decision are therefore the framework within which every technique in the chapter sits.*

CLASSIFICATION OF UROGENITAL SINUS AND CLOACAL ANOMALIES

- A **urogenital sinus** is a persistent communication of the vagina with the urinary tract. It can lie anywhere from meatus to bladder but is **most often in the mid to distal urethra**; the two structures exit the perineum as one common channel.

THE FOUR ENTITIES WITH A PERSISTENT UROGENITAL SINUS

Entity	Distinguishing feature
Genital atypia states	Commonest; most frequently CAH (Fig. 60.1)
Pure urogenital sinus	Normal external genitalia (Fig. 60.2)
Persistent cloaca	Rectum also involved: genital, urinary and intestinal tracts exit through one perineal opening
Female exstrophy	Thought to represent a form of persistent urogenital sinus

- The table defines the territory of the chapter. The trap is to equate "urogenital sinus" with CAH alone: the pure urogenital sinus behaves anatomically more like a cloaca.
- **Descriptive bias by specialty:** urologists described the vagina entering the urethra; gynaecologists described the urethra entering the vestibule.

- The external cloacal orifice looks **vaginal in some patients and like a near-normal urogenital orifice in others**. Figures show a urethral-type sinus, sometimes with an anteriorly placed rectum (Fig. 60.3), and a vaginal-type sinus with high or low confluence (Fig. 60.4).
- **Governing principle: the confluence position relative to the bladder neck (true urethral length) matters more for surgery than common-channel length.** The same holds in cloaca, where it now directs management.

KEY POINTS: UROGENITAL SINUS ANOMALIES — GENERAL

- Urogenital sinus anomalies occur in four entities: genital atypia states, pure urogenital sinus, persistent cloaca with rectal involvement, and female exstrophy.

Concept — *Whatever the label or external appearance, **true urethral length** (bladder neck to confluence) is the variable that decides the operation, in urogenital sinus and cloaca alike.*

Evaluation

Congenital adrenal hyperplasia

- **Urogenital sinus is most often seen in DSD states, chiefly CAH;** the nonclassic mild form affects up to 1 in 500.
- **CAH is the commonest cause of atypical genitalia in newborns:** half of all cases and 90% of those with a 46,XX karyotype.
- CAH is a group of **autosomal recessive** enzyme defects. The commonest causing genital atypia is **21-hydroxylase deficiency**, about 1 in 15,000–16,000. The gene is **CYP21A2 on chromosome 6p**, accounting for 90–95% of CAH; virilisation corresponds to the mutation type.
- CAH is a **spectrum disorder**, from mild to life-altering. The first priority is **accurate diagnosis**, with the karyotype **within 48 hours**.
- **Salt wasting affects 67–75%** through aldosterone deficiency, so fluid and electrolytes must be monitored.
 - **Classic findings:** hyponatraemia, hyperkalaemia, high urinary sodium, low serum and urinary aldosterone, raised plasma renin activity.
 - Consequences: dehydration, poor feeding and hypotension, progressing to vascular collapse, shock and death (**adrenal crisis**).
- **Work-up:** rapid karyotype on cultured peripheral leucocytes, serum electrolytes and CAH hormone levels; results shared with the team and family.
- Gender is increasingly understood as fluid rather than simply binary. **The child is not named until evaluation is complete** and the family has met the team.

Cloacal anomalies

- **Far more complex and multisystem, but rare:** about 1 in 20,000–50,000.
- The **most challenging anorectal malformations (ARMs)**, making up about 5–13.6% of ARM.
- External genitalia vary widely; a prominent phallus-like structure may prompt an **initial DSD evaluation**.

Concept — *In the newborn with a urogenital sinus, the **first threat is metabolic, not anatomical**: diagnose CAH and anticipate adrenal crisis before any thought of reconstruction.*

History and Physical Examination

History

- **Antenatal ultrasonography** identifies many cases: fluid-filled pelvic structures (vagina and bladder) with indeterminate genitalia. **Newborn CAH screening** has run in all 50 US states since 2009 and in many regions worldwide.
- History and examination often reach the diagnosis. Ask about:
 - **maternal medication in pregnancy**, especially androgens;
 - **family history of early infant death or fluid and electrolyte disturbance**, which suggests CAH;
 - siblings with genital atypia or gender dysphoria;
 - relatives with abnormal puberty.

General and abdominal examination

- **Assess overall health before the genitalia**; abnormal facies suggest other syndromes.
- **Hypertension points to CAH from 11 β -hydroxylase deficiency**, so blood pressure must be recorded. Dehydration also suggests CAH.
- A **suprapubic mass** may be a distended bladder, **hydrometrocolpos**, or both. Hydrometrocolpos is frequently the **initial sign** and often the **only early finding in pure urogenital sinus**.

Hydrocolpos in cloaca

- Abdominal distension may be **severe**, combining hydrometrocolpos, bladder and bowel distension. Hydrocolpos was common in early series, although its frequency varied widely, and is usually accompanied by **uterovaginal duplication**.
- **It is more frequent with** a common channel over 3 cm, vaginal duplication, posterior cloaca, and a rectocloacal rather than rectovaginal communication.
- **Mechanism**: urine flows preferentially through the sinus into the vagina(s) on voiding, and the vagina drains poorly. **Maternal oestrogen** drives cervical mucus secretion, adding to the distension.
- **Urine entering the rectum** can cause significant fluid and electrolyte disturbance.
- Severe pelvic distension can cause **oedematous, cyanotic legs, respiratory distress and acidosis**.

Spine

- **Spinal cord anomalies** are associated with urogenital sinus and are **very common with cloaca**.
- Cutaneous markers are a sacral dimple, hair patch or abnormal pigmentation. **More often the clue is bony**: an abnormal buttock crease or flat buttocks from **sacral agenesis**.

Genital examination

- Record clitoral size, the consistency of the erectile bodies, and any curvature.
- **Huffman clitoral index** = glans width $\times$ clitoral length: **normal below 3.5 mm²; above 10 mm² is of concern**. A term female clitoris is about **4 mm** long and **3.3 mm** wide.
- **Persistent isolated clitoral hypertrophy in preterm infants is not a DSD**.
- **Meatus of the sinus**: anywhere from a near-normal introital position to the tip of a well-formed glans.
- **Gonads**: record number, position and consistency. **If both gonads are descended, a 46,XX karyotype is extremely unlikely**.
- **Labioscrotal folds**: note their relation to the clitoris and rectum and the degree of fusion. **Hyperpigmentation of the folds and areolae** in CAH reflects raised melanocyte-stimulating hormone.
- **Anus**: usually normally sited, but **anterior displacement is not uncommon** and bridges to cloaca. The pure urogenital sinus aligns more with cloaca than with the DSD urogenital sinus.

- **Posterior cloaca:** the sinus opens into the anus (type A) or just anterior to it (type B).
- A gentle **rectal examination** may identify a palpable cervix.

Examination of the persistent cloaca

- There is a **single perineal orifice**, because the rectum also enters the common channel.
- External appearance ranges from near-normal female to **complete genital transposition** or a **blank, "doll-like" perineum**, with vaginal-type and urethral-type appearances (Fig. 60.5).
- An enlarged phallus-like structure may be present. The opening may exit through an apparently normal introitus or at the tip of the phallus-like structure, and a **secondary accessory channel** at the tip has been reported.

ASSOCIATED ANOMALIES IN PERSISTENT CLOACA

System	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, duplication, crossed fused ectopia; bladder duplication; anomalies of number, position, rotation and echogenicity	Majority
Uterovaginal	Duplication and septation of vagina and uterus	53–67%; septation about 60%
Vertebral, especially sacral	Sacral anomalies, spinal dysraphism	Common
Cardiovascular, central nervous system, respiratory	Structural anomalies	Minority
VACTERL association	Vertebral anomalies, anal atresia, cardiac anomalies, tracheo-oesophageal fistula and/or oesophageal atresia, renal agenesis and dysplasia, limb defects	Tracheo-oesophageal fistula often present
Gastrointestinal	Duodenal atresia, rectal duplication	Rare
Malrotation	Same frequency as the general population in ARM, commoner in severe malformations	Correct at colostomy

- The table is a **screening checklist**: kidneys, Müllerian structures and spine are almost expected to be abnormal.
- Malrotation found at colostomy is corrected then, and the **appendix is preserved** for a later appendicocaecostomy for antegrade enemas.

KEY POINTS: UROGENITAL SINUS ANOMALIES — EVALUATION

- Key findings on history and examination: family history of sudden infant death, fluid and electrolyte abnormality, hypertension, suprapubic mass, lower-limb cyanosis, palpable gonads, hyperpigmentation, and the number of perineal orifices.
- Identify the level of the urethrovaginal confluence and the position of the rectum radiographically or endoscopically, relative to the bladder neck. Ultrasonography is mandatory for the uterus, ovaries and kidneys.

Concept — The examination separates **two early dangers: adrenal crisis (CAH) and obstructive hydrocolpos** (pure urogenital sinus and cloaca). The number of perineal orifices tells you which world you are in.

Radiographic and Endoscopic Evaluation

Urogenital Sinus Abnormalities

- **Define before surgery:** common-sinus length, confluence site and its distance from the bladder neck, vaginal size and number, whether a cervix is present, and bladder and urethral anatomy, radiographically or endoscopically.
- **The critical determinant of the operation is the vagina's position relative to the bladder neck**, far more than common-channel length. The same concept produced the new cloacal algorithm (Wood et al., 2017).
- The confluence lies on a **continuum from bladder to near-normal perineal position** (Fig. 60.6). It is not a simple choice between "high" (proximal, suprasphincteric) and "low" (distal, infrasphincteric).
- **Ganesan concept: in CAH the urethrovaginal confluence is constant, with a normal proximal urethra.** Bladder neck to confluence is generally **1.5–2 cm in CAH infants**. What lengthens with virilisation is the **vagina-to-meatus distance**, so a "high" confluence in CAH is a product of virilisation.
- **The illusion of height:** a short common channel looks low and a long one looks high. In the virilised sinus this is reinforced by a male-like external sphincter, a verumontanum-like confluence just proximal to it, and a masculinised bony pelvis.
- The **Rink classification** records exact common-channel length and bladder-neck-to-vagina distance, with confluence level, clitoral size and external appearance.
- **Ultrasonography is mandatory:** kidneys, ovaries, uterus, and any bladder or vaginal distension. **Enlarged, cerebriform adrenals indicate CAH.**
- **Genitography** fills bladder, urethra, vagina and sinus with contrast (Figs. 60.7–60.8).
Foley balloon occluding the perineal meatus → contrast injected → catheter into the bladder as for a voiding cystourethrogram → images on voiding (some vaginas fill only then).
A **cervical impression at the vaginal dome** indicates normal female internal organs.
- **Magnetic resonance imaging (MRI)** is of limited value in CAH but very helpful in pure urogenital sinus, other DSDs and cloaca.
- **In CAH, endoscopy at reconstruction nearly always defines the anatomy**, so genitography is less useful: in the authors' series it was often inaccurate, added nothing beyond endoscopy and did not change the type or timing of surgery.
- **Endoscopy is the most helpful study for surgical planning.** In CAH it is done at reconstruction. It is a **separate early procedure** for gender assignment in other DSDs, complex pure urogenital sinus, or when genitography fails to show the vagina.
- If the vagina is not identified, look for **multiple punctate openings in the proximal urethra**; pass a ureteral catheter into the opening and study the vagina fluoroscopically. Record vaginal position, size and number (Fig. 60.9: the urethra is superior and the vagina inferior).
- In severe masculinisation there may be a **male-like external sphincter** with the vagina entering just proximal to it in a verumontanum-like structure, and occasionally a completely masculinised urethra.
- **Gonadal biopsy or inspection of the internal genitalia** is rarely needed; it is now mostly laparoscopic.
 - Perform it **only if the result would change the gender of rearing.**
 - Use a **deep longitudinal incision**, because the ovarian component of an ovotestis may completely surround the testicular component or lie at the poles.
- All children with a DSD undergo chromosomal, endocrine and genetic evaluation (Chapter 59).

IMAGING AND ENDOSCOPY BY CONDITION: CAH VERSUS PURE UROGENITAL SINUS AND CLOACA

Modality	CAH urogenital sinus	Pure urogenital sinus / cloaca
Ultrasonography	Mandatory; cerebriiform adrenals	Mandatory for renal anomalies and hydronephrosis; pelvic ultrasonography poor for Müllerian anatomy
Genitography	Least useful ; may be avoided	Mandatory, now as a CT or MRI cloacogram
MRI	Limited value	Very helpful: pelvis, spine, sphincter muscles
Endoscopy	At reconstruction; defines anatomy	Early, separate procedure to decompress and define anatomy

- The pattern: **CAH anatomy is predictable** (constant proximal urethra), so it needs less imaging; the less predictable the anatomy, the more modalities are required.

Cloacal Anomalies*Prenatal diagnosis*

- Persistent cloaca is increasingly diagnosed prenatally, **as early as 19 weeks**.
- **Diagnostic combination:** transient fetal ascites + bilobed or trilobed pelvic cysts + bilateral hydronephrosis + reduced amniotic fluid.
- **Mechanism of the ascites:** vaginal outlet obstruction → urine refluxes through the uterus → out of the fallopian tubes into the peritoneum.

Postnatal imaging

- **Begin with a plain abdominal film and abdominal ultrasonography.** A pelvic mass may be obvious on a kidneys, ureters and bladder film.
 - Retrograde flow of urine and meconium produces the **classic linear calcifications or calcified meconium**.
 - Granular calcification along the rectum reflects urine in the rectum.
- **Pelvic ultrasonography is of limited use for Müllerian anatomy.**
- **Abdominal ultrasonography** shows renal anatomy and **hydronephrosis, which is common**. It usually results from **hydrocolpos compressing the bladder neck**, sometimes from ureteric compression or primary obstructive megaureter. Renal anomalies (dysplasia, fusion, ectopia, pelviureteric junction obstruction, duplication) are reported in **25–83%**.

Cloacogram and endoscopy

- Unlike CAH, **genitography and endoscopy are mandatory in cloaca**. Most now use a **computed tomography (CT) or MRI cloacogram combined with endoscopy**.
General anaesthesia → endoscopy → balloon catheters inflated in bladder, vagina(s) and rectum → contrast through these and through a mucous-fistula catheter at CT or MRI → three-dimensional (3D) reconstruction.
- **No modality is perfect:** some hold endoscopy best, others 3D reconstructed imaging.
- **Targets:** rectal and vaginal confluences; **urethral length** and its communication with the cloaca; vaginal anatomy; position of the **true rectum**.
- The urinary communication is usually urethral, but in a substantial minority there is **virtually no urethra** and it lies at bladder-neck level.

Vaginal and uterine anatomy

- **Cloaca impairs Müllerian duct fusion**. Single and duplicated vaginas are about equally common, the vagina is absent in a minority, and in modern series **most have vaginal and uterine duplication** (Fig. 60.10: duplicated fluid-filled vaginas behind the decompressed bladder).

- Preoperative definition succeeds best with **vaginography** and worst with **pelvic ultrasonography**; cloacograms sometimes disagree with operative findings.
- **Duplication patterns**: usually side by side with a single opening into the cloaca, sometimes separate openings. The vaginas may differ in size, one may enter the sidewall of the other, and **a cervix usually tops each vagina**. Vaginal entry lies on a spectrum from bladder to near the perineum.
- **The uterus usually mirrors the vagina** (two vaginas, two uteri). The vagina may be absent with the uterus present, or the uterus hypoplastic. Bicornuate, hypoplastic or didelphic uterus is found in about a third.

Rectal confluence and common channel

- **Rectal confluence** is most often at the level of the vaginal confluence. It may be broad or a long narrow fistula, and may enter the vagina or bladder without reaching the cloaca. The commonest site is **within the septum of a duplicated vagina**. The communication is usually vaginal, less often cloacal.
- **The true rectum may lie well above the fistula**; locating it matters as much as locating the fistula.
- **Peña's categorisation by common-channel length (under or over 3 cm)** is now almost universal.
- **Channel type**: narrow and urethra-like, or large, redundant and vagina-like; roughly half each.

COMMON CHANNEL: SHORT VERSUS LONG, URETHRAL VERSUS VAGINAL TYPE

	Common channel under 3 cm	Common channel over 3 cm
Associated anomalies	Fewer	More (urological defects in nearly all)
Hydrocolpos	Less frequent	More frequent
Abdominal approach	Rarely needed	More likely
Continence	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often
	Urethral-type channel	Vaginal-type channel
Appearance	Narrow, like a urethra	Large, redundant, like a vagina
Consequence	Higher outlet resistance → hydrocolpos	Lower resistance

Associated-system imaging and early endoscopy

- **Echocardiography always. MRI is necessary** for the lumbosacral spine and for pelvic anatomy and musculature.
- **Spine**:
 - sacral anomalies are common, and MRI detects spinal cord anomalies in up to 43%;
 - about **one-third have a tethered cord**, yet **skin stigmata are absent in most** of them;
 - MRI also defines the level of rectal atresia and sphincter development.
- **Early separate endoscopy** is often needed to decompress the vagina and bladder and define the anatomy.
 - The vagina is easy to see, but **bladder entry can be very difficult or impossible in a neonate** because the distended vagina compresses the bladder anteriorly. **Empty the vagina first.**
 - The rectal fistula can be hard to find. Irrigate mucus and faeces through the scope and the mucous fistula; a small catheter through the mucous fistula localises the rectal confluence.
 - Document the urethral length to each confluence and the **common-channel length, which predicts faecal continence.**

Concept — Cloacal imaging is organised around **three measurements: common-channel length, true urethral length and the levels of the rectal and vaginal confluences**. The reconstruction is chosen from these, and the spine and kidneys set the prognosis.

SURGICAL RECONSTRUCTION OF DISORDERS OF SEX DEVELOPMENT AND UROGENITAL SINUS

Initial Management, Timing, and Principles

The case for and against early surgery

- **Most children with a urogenital sinus have genital atypia** and are first assessed by the multidisciplinary DSD team; the family must know **all the controversies**.
- **Historical rationale for early "normalising" surgery:** positive psychosocial adjustment; relief of parental distress (limited evidence); the belief that raising a child with atypia is very difficult; parental stress at "doing nothing". Surgery was advised at **2–6 months**, when tissues are most pliable and psychological trauma minimal.
- **Core problem:** assigning gender when the individual's eventual **gender identity is unknown**.
- **Opposition:** InterACT, Human Rights Watch, the United Nations, three former US Surgeons General and some physician groups question early surgery. Most support a **moratorium until the patient can express a wish**, at an age not yet defined.
 - Informed consent is impossible in an infant, and many regard genital surgery as the patient's right.
 - The ethics of parental decision-making remain unresolved.
 - The infant is the passive centre of the team yet ultimately has the most important voice.
 - Data on the impact of raising a child with atypia are lacking.
- **Counselling standard:** present pros and cons without bias. **Observation with psychological and peer support must be an option**, and the same support is owed to families who choose early surgery.

What surgery can and cannot achieve

- Excellent cosmesis is achievable, but **sexual identity is multifactorial** and the **brain is the dominant organ in sexual orientation**.
- Little is known of adult adjustment after surgery, or of the path an operated infant would have chosen as an adult.
- Outcomes of infant surgery are not known for about **20 years**, so published results may describe abandoned techniques.
- **Genital surgery does not "cure" a DSD.** Females with CAH are **less satisfied with their genitalia than controls, whether operated or not**. Women with unoperated clitoromegaly from CAH report **no net benefit** from it and harm across social, romantic and emotional domains.

Counselling and parental decision-making

- There is no absolutely correct answer. **Never withhold information; secrecy has no place**; counselling is unbiased and transparent.
- At the authors' institution, **all parents and patients meet a clinical psychologist before surgery**, and families are directed to all major advocacy groups whatever their stance.
- Attitudes to early surgery, especially clitoral surgery, have shifted.
- **Parental priorities:** the largest concerns were **not surgical** but the life-threatening and developmental effects of CAH and medication side effects. The child's non-involvement in the decision ranked among the least prioritised.
- **Decisional regret after genital surgery** is mild to moderate in a minority of parents and severe in none. It is lower than after hypospadias repair, adenotonsillectomy or paediatric cancer treatment, and **no parent preferred delayed surgery**.

- Hereafter the chapter assumes all parties agree that surgery is warranted. **The techniques apply from infancy to adulthood.**

Anatomy to be corrected

- Clitoral hypertrophy; fusion and anterior displacement of the labia majora; **absent labia minora**; a common urogenital sinus. Fig. 60.11 shows moderate virilisation with incomplete or complete labioscrotal fusion, and severe virilisation.
- Normal clitoral, vaginal and labial size varies considerably.
- **Three historical steps:** clitoroplasty → labioplasty → vaginoplasty.

Timing of clitoroplasty

No consensus in the mid-1900s → about **1 year** → **first few months** (1980s) → **1–3 weeks** (1990s).

- Parents now less often choose surgery for minimal hypertrophy.
- **Current indication: severe hypertrophy only (Prader III–IV)**, the basis of recent support for early surgery (see consensus table).
- Satisfaction with clitoral function is **higher the younger the age at surgery**.
- **Clitoral surgery, whether recession or resection, is the most controversial component** of the whole debate.

Timing of vaginoplasty

- **The vagina has no function in childhood**, and some base timing partly on confluence level.
- **Low (distal) confluence:** simultaneous clitoroplasty, vaginoplasty and labioplasty is the historical standard. **High confluence:** see table below.
- **US practice:** in a national paediatric hospital database, genital surgery was done at a mean of about 10 months. Vaginal and perineal surgery predominated; clitoral surgery was done in about half.
- **Patient and parent views:** most CAH patients and parents **support the parents' right to decide and oppose moratoria**. They are satisfied with early surgery, and **those operated later generally wish it had been earlier**.
- **There is no universal agreement on vaginoplasty timing and no adequate data comparing early with late surgery.**

TIMING OF VAGINOPLASTY FOR HIGH CONFLUENCE: THE THREE POSITIONS

Position	Arguments
Early single-stage (with clitoroplasty and labioplasty)	Redundant clitoral skin and sinus tissue available, but compromised once mobilised. Maternal oestrogen gives thicker, better-vascularised vaginal tissue. Planned one-stage repair gives better results. Stenosis is higher after pubertal than infant surgery
Delay until after puberty	High stenosis rate after infant surgery; avoids dilation; the patient can consent. Delay the small hypoplastic vagina (under 3 cm) . Pubertal oestrogen, vaginal growth and distension may improve healing and outcome
Hybrid	Clitoroplasty and labioplasty in infancy; vaginoplasty at puberty , especially for high confluence
Rebuttal of pubertal surgery	Surgery is not easier at puberty and there is no evidence healing is better

- **Oestrogen is invoked by both camps** (maternal oestrogen in infancy, endogenous oestrogen at puberty). The trap is to cite it as favouring only one side.

CONSENSUS, GUIDELINE AND PROFESSIONAL POSITIONS ON DSD GENITAL SURGERY

Body (year)	Position
Joint Lawson Wilkins Pediatric Endocrine Society (LWPES) / European Society for Paediatric Endocrinology (ESPE) CAH Working Group (2002)	Surgery at 2–6 months (stated in the chapter for high confluence); surgery between 12 months and adolescence not recommended
Consensus statement (2006)	Surgery only for Prader III–IV ; evidence inadequate to abandon early separation of vagina and urethra
Timing review (2010)	Consider clitoral surgery in infancy for Prader III or greater
International Society of Hypospadias and DSD, Fourth World Congress poll	Most surgeons prefer surgery before 2 years , single-stage
Endocrine Society CAH guideline (2018)	Surgical decisions are the prerogative of families , in joint decision-making with experienced surgeons
American Medical Association Council on Ethical and Judicial Affairs (CEJA)	After 2 years' review, rejected a moratorium
Advocacy groups; some US state bills	Moratorium until the patient can consent. Parental decisions viewed as a human-rights violation ; the bills were proposed but not enacted

- The **ultimate decision currently rests with the parents in concert with the multidisciplinary team**, presented with a balanced account of all risks.

Clitoroplasty: evolution of technique

- The **clitoris is a sexual organ**. Clitoroplasty aims at excellent cosmesis **and preserved innervation**.

EVOLUTION OF CLITORAL SURGERY

Eponym / era	Technique	Verdict
Young (1934)	Glans preserved on a flap	Glans sloughed
Amputation era	Complete excision of clitoral tissue, to avoid painful erection	Based on the claim that the clitoris is unnecessary for sexual response; abandoned
Lattimer (1961)	Recession into subcutaneous fat; glans denuded	Glans denudation not recommended
Kaplan	Corpora split; transverse Heineke–Mikulicz closure	Recession
Pellerin; Randolph and Hung	Corpora buried beneath the pubis	Recession
Schmid (1961)	First excision of corporeal tissue preserving the neurovascular bundle, glans intact	Basis of virtually all later techniques ; glans usually well, occasional shrinkage or devascularisation
Spence and Allen	Whole shaft excised; glans kept on the ventral urethral plate	Sacrificed the entire neurovascular bundle
Kogan (1983)	Subtunical excision through lateral incisions in Buck fascia	—
Baskin (1999)	Neurovascular anatomy defined (Fig. 60.12)	A ventral (not lateral) incision spares the dorsal bundle and its laterally fanning branches
Pippi Salle (2007)	Corpora disassembled and placed in the labial fat	Potentially usable for a phallus if male identity emerges; results unknown

- The arc runs **amputation → recession → nerve-preserving excision → ventral subtunical excision**, each step tracking better knowledge of clitoral innervation. The authors hold that **Buck fascia with its**

neurovascular bundle, and the glans, must always be preserved. Research into clitoral neuroanatomy and sensory testing continues.

Vaginoplasty and labioplasty: foundations

- **Almost every vaginal repair uses a posteriorly based perineal flap:** proposed by Lattimer and described by **Fortunoff (1964)**. The wide-based **Fortunoff flap** has been modified to an **omega-shaped flap** with better cosmesis.
- **Hendren and Crawford (1969)** introduced the **pull-through vaginoplasty** for high confluence. Their principle that **confluence level determines vaginoplasty type** underlies all vaginoplasty today.

THE FOUR TYPES OF VAGINOPLASTY

Type	Indication	Principle	Pitfall
Cut-back	Simple labial fusion only (some use it for Prader I–II)	Incision of the sinus	Rarely used; does not reach the normal-calibre vagina
Flap	Low (distal) confluence	Posterior sinus and vaginal walls opened, anterior vaginal wall left intact, perineal flap inset. Confluence level unchanged: the introitus and sinus are only widened	Mild female hypospadias common. Never for a very high confluence: short hypospadiac urethra, vaginal voiding, infection, incontinence
Pull-through	Any level; reserved for very high confluence	Vagina separated from the sinus; the sinus becomes the urethra	The vagina often fails to reach the perineum, so skin flaps are needed
Complete replacement	Rudimentary or absent vagina only	Several techniques	—

- The organising question is whether the **confluence must be moved**. A flap brings skin up to a low vagina; a pull-through takes a high vagina off the urethra. **Applying a flap to a high confluence is the classic error.**
- **Labioplasty:** the labia minora are absent and the labia majora lie above the new introitus. Labia minora are made from **split clitoral skin (Marberger)**, and both labia are moved **inferiorly by YV-plasty**, so the vagina sits between them.
- In Fig. 60.13, a skin cuff around the glans is kept for the clitoral hood. The preputial skin is split for the labia minora, and the proximal preputial skin is plicated on its undersurface to create a "hooded" appearance.

Concept — Early surgery offers **technical advantages** (untouched tissue, maternal oestrogen, single stage); delay offers **autonomy** (consent, known identity). With no comparative data on either side, the **decision is ethical rather than evidential**.

Current Operative Techniques for Female Disorders of Sex Development and Urogenital Sinus Repair

Preoperative preparation and positioning

Metabolic stability (especially CAH) → bowel preparation usually unnecessary; complete polyethylene glycol–electrolyte (GoLYTELY) preparation if a high confluence is suspected → **broad-spectrum antibiotics** for all → **"stress-dose" steroids in CAH** → general anaesthesia → endoscopy → **Fogarty catheter into the vagina**, balloon inflated, clamped, left indwelling → **Foley catheter in the bladder** → both kept sterile → skin preparation (povidone-iodine or chlorhexidine).

- **Lithotomy** is common but restricts the view to the surgeon, hinders teaching, and prevents conversion to a posterior or abdominal approach.

- **Authors' method: total lower-body preparation from nipples to feet** in every prepubertal child, with legs wrapped and the lower body passed through the drape. This gives access to perineum and abdomen and allows **rotation between supine and prone** (Fig. 60.14). Postpubertal patients often need lithotomy.

KEY POINTS: UROGENITAL SINUS ANOMALIES — RECONSTRUCTION

- The three historical steps are clitoroplasty, vaginoplasty and labioplasty.
- Clitoroplasty is very controversial in infants. When performed, preserve the glans and tunics with their neurovascular bundles; excise erectile tissue from the ventral aspect only.
- Four vaginoplasty types: cut-back (generally no longer used); flap for low-to-mid confluence; pull-through for high confluence; replacement for a rudimentary or absent vagina. Timing is controversial.
- Labia minora are absent and are reconstructed from clitoral skin; the anteriorly placed labia majora are moved inferiorly.
- A Fogarty catheter with its balloon inflated in the vagina is helpful. For low confluence, an omega-shaped perineal flap gives better cosmesis; it is sewn into the posteriorly opened sinus and must reach the normal-calibre vagina.
- For high confluence, the vagina may need separation from the urinary tract and "pull-through" towards the perineum, with the sinus closed as a urethra. For a very small, very high vagina, strongly consider waiting until puberty.

Low Vaginal Confluence: Clitoral Hypertrophy

- **Most children operated on for DSD or urogenital sinus have a low confluence suited to flap vaginoplasty**, usually with clitoral hypertrophy. Clitoroplasty follows full counselling; the techniques apply at **any age**.

Marking

Supine → traction suture through the glans → incisions marked → optional subcutaneous **0.5% lidocaine with 1:200,000 adrenaline** → incision around the glans, **leaving the entire inner prepuce intact** → parallel lines either side of the **ventral mucosal strip (urethral plate equivalent)**, around the meatus → **omega-shaped perineal flap**, apex near the meatus → **Y-shaped incision** around the inferior aspect of each labium majus.

Degloving and anatomy

- The incision begins **dorsally**. Inner preputial skin is kept for the clitoral hood; it is **second only to the glans in sensitivity**.
- Deglove while **keeping the ventral "urethral plate" intact with the meatus**, to the corporeal bifurcation ventrally and the pubis dorsally, without injuring neurovascular tissue.
- **Arterial supply**: internal pudendal artery from **Alcock canal** near the ischial tuberosity → runs ventrally on the **medial aspect of the bifurcated corpora** → then dorsally along the shaft.
- **Nerves**: ascend along the **ischiopubic rami** → join as paired bundles on the **dorsal surface** → pass largely intact into the glans. **Whatever the technique, this innervation must not be disturbed.**

Erectile tissue excision (authors' technique)

Tourniquet at the clitoral base, or corpora compressed against the pubis with Kittner dissectors → **longitudinal incisions through Buck fascia on the ventral-most aspect** of each corporal body, glans to bifurcation (Fig. 60.15) → erectile tissue dissected (teased) out of the tunics → **tunics left undisturbed; no tunical tissue excised** → **dorsal neurovascular bundle not mobilised or disturbed** → proximal ends of erectile tissue suture-ligated.

CLITOROPLASTY TECHNIQUES COMPARED

Technique	Principle	Comment
Ventral subtunical excision (authors)	Erectile tissue removed through ventral incisions; tunics and dorsal bundle untouched	Authors' standard
Neurovascular bundle mobilisation	Bundle lifted off the corpora	Authors consider it more likely to injure sensitivity and vascularity . If done, dissect below the second layer of Buck fascia, just beneath the tunica albuginea
Reapproximation	Intervening erectile tissue excised; distal corpora joined to proximal	—
Corporal preservation	Corpora split in the midline, mobilised laterally and fixed to the pubic bone; clitoris shortened without removing tissue	No long-term data
Disassembly (Pippi Salle)	Corpora dissected off the bundle, separated and buried beneath the labia	Reversible if later gender dysphoria

The glans

- **Glans reduction is controversial.** If attempted, any excision is **ventral, near the midline**, like hypospadias glans wings, because the glans is innervated by **perforating branches at the dorsal glans-corporal junction**.
- **Do not excise glanular epithelium**, since the sensory neuropeptides lie just beneath it. **No data show that a large glans or clitoris harms sexual function.**
- The glans is secured to the **coronal stumps**; a glans sewn to the pubis lies abnormally high and prominent.

Flap vaginoplasty

Omega flap incised with its underlying fat → sinus exposed → posterior sinus and vagina dissected off the rectum (**the most difficult step**) → stay sutures in the meatus → **posterior sinus wall opened in the midline into the posterior vaginal wall** → continued **until normal-calibre vagina is reached** (the distal third is usually narrowed) → interrupted sutures from flap to split posterior vaginal wall.

- **The flap must be** long enough for a **tension-free** anastomosis, wide enough for a normal-calibre introitus **without compromising the perineal-body blood supply**, and **not redundant**, which would form an obstructing introital mound.

Labioplasty

Clitoral skin unfurled and divided in the midline, stopping well short of the base → base combined with preputial skin as the **clitoral hood** → split skin brought down as **labia minora**, sutured to the ventral plate and lateral vaginal wall → Y-incisions around the labia majora → labia majora mobilised and fixed inferiorly beside the vagina (**YV-plasty**).

- Inferior placement often gives an **M-shaped hood**; **plicating the dermis** restores an inverted U. The end point is an introitus lying **between the labia minora and majora**, not an isolated perineal hole.

Concept — *The low-confluence repair has two fixed points: **never touch the dorsal neurovascular bundle**, and **always open back to normal-calibre vagina**. Most historical failures broke one of these.*

High Vaginal Confluence: With or Without Clitoral Hypertrophy

- **Complete separation with pull-through vaginoplasty** (Hendren and Crawford) is the accepted solution; some reserve it for a **sinus longer than 3 cm**.

- High confluence occurs in only **about 5% of CAH** and is **more common in pure urogenital sinus**. In severely virilised CAH, apparent height reflects a **lengthened common sinus plus a male-like sphincter**, with minimal urethral shortening.
- **The original pull-through** left an **isolated vaginal opening** separate from the rest of the genitalia and **without mucosal lining**, and gave poor vision at the critical points.
- **Refinements:**
 - **Passerini-Glazel flap:** the mobilised sinus is divided dorsally, tubularised with clitoral skin and folded back towards the vagina. This improves cosmesis, covers the separation and helps prevent urethrovaginal fistula; it was later used as a flap for the **anterior vaginal wall**.
 - **Gonzalez-Fernandes preputial flap:** preputial skin builds the vestibule and anterior wall.
- **The critical step is separating the anterior vaginal wall from the urethra and bladder neck.** There is no obvious plane, the sphincter and a rich nerve supply are at risk, and it is the hardest area to see. Poor exposure leads to **stricture, fistula, diverticulum or retained distal vagina**.

APPROACHES TO EXPOSURE OF THE HIGH CONFLUENCE

Approach	Features	Status
Supine or lithotomy	Conventional	Most surgeons
Transtrigonal mobilisation	Vagina mobilised through the trigone	Seldom necessary
Anterior sagittal transanorectal approach (ASTRA)	Prone; anterior rectal wall divided in the midline	Excellent exposure for very high confluence; no colostomy ; continence unaffected. Rarely needed in CAH, useful in high pure urogenital sinus
Lateral rectal mobilisation	Rectum displaced laterally	Abandoned as too difficult
Posterior prone midline approach (authors)	Rectum retracted, not divided	Others have not found it necessary

Pull-through technique (prone approach)

Endoscopy, catheters, total-body preparation → **clitoroplasty supine** if needed → **rotate prone** (pure urogenital sinus starts prone) → omega flap incised and retracted posteriorly → **midline dissection between posterior sinus/vagina and rectum**, rectum retracted with a small Deaver retractor (Fig. 60.16) → sinus divided in the posterior midline to normal-calibre vagina → Deaver placed **inside the vagina**, retracted upwards → anterior vaginal wall dissected off the urethra **under direct vision** (Fig. 60.17) → sinus **tubularised as the urethra in two or three layers over a Foley** → anterior vagina mobilised towards the perineum.

- At the separation the tissues are thin: **always err on the side of the vagina**. Dissection becomes easier proximally.
- **"Pull-through" is often a misnomer:** the vagina frequently falls short, and **flaps are brought up to it**. Options are preputial skin to the spatulated anterior vagina (Gonzalez-Fernandes flap, Fig. 60.18), or, without preputial skin, a **buttock or laterally based skin flap** (Fig. 60.19).
- Then supine → posterior perineal flap anastomosed as for flap vaginoplasty → labioplasty. If a laterally based flap formed the anterior wall, **relocation of the labia majora is deferred**.
- **Skin flaps are best avoided** (stenosis, hair); **urogenital sinus tissue is more reliable**. After multiple flaps, the **legs are loosely bound together** postoperatively.

Concept — High-confluence surgery is dominated by **one dissection: the anterior vaginal wall off the urethra and bladder neck**. Prone positioning, ASTRA, sinus flaps and urogenital mobilisation are all ways to see that plane better or to avoid it.

Total and Partial Urogenital Mobilization

- **Peña (1997): total urogenital mobilisation (TUM)** was devised to complete the sinus component of cloacal repair. **The entire sinus is dissected circumferentially and moved towards the perineum.** Peña reported **operative time cut by about 70%**, superior cosmesis, and **less fistula, vaginal stenosis and acquired atresia**, because vaginal separation is usually unnecessary.
- TUM has since been applied to urogenital sinus, female exstrophy and penile agenesis.
- **Uses of the mobilised sinus:**
 - amputated, with vagina and urethra sewn flush;
 - **mucosa-lined vestibule;**
 - **Passerini-type flap** for the anterior vaginal wall in a pull-through;
 - **replacement of the posterior skin flap**, either split laterally and rotated posteriorly, or halved longitudinally with both halves rotated inferiorly.
- **The authors incorporate mobilisation into nearly every urogenital sinus repair.** A mid-level confluence reaches the perineum without vaginal separation; the highest still need a pull-through, which is **much easier after mobilisation.**
- **TUM is not a vaginoplasty type in its own right** but enables one, and a flap or pull-through may still be needed. The vagina is **rarely sewn flush**, because a circumferential anastomosis risks stenosis.

Technique

Endoscopy (confluence level; Fogarty in vagina, Foley in bladder) → sponge in rectum → incisions marked, optional lidocaine with adrenaline → clitoroplasty, with sinus meatus and urethral plate circumscribed by stay sutures → sinus mobilised off the corpora to the bifurcation → **midline dissection behind the sinus to the peritoneal reflection**, exposing the whole posterior vaginal wall (Figs. 60.20–60.21) → **circumferential mobilisation directly on the sinus, beneath the pubis** → avascular pubic ligaments divided → the sinus "gives" and moves towards the perineum (Fig. 60.22) → Fogarty palpable → posterior vaginal wall opened between stay sutures.

- **Vagina now near the perineum:** sew it flush, or (preferred) inset an **omega flap into the spatulated posterior wall; split the sinus ventrally** for a mucosa-lined vestibule (Fig. 60.23).
- **Vagina still high:** separate the anterior wall from urethra and bladder neck (most easily prone), close the urethral opening in two layers, and **split the sinus dorsally** as a Passerini-type flap for the anterior wall (Fig. 60.24). Posterior flap and labioplasty follow.

HOW THE MOBILISED SINUS IS SPLIT (FIG. 60.24)

Flap vaginoplasty	Pull-through vaginoplasty
Sinus split ventrally	Sinus split dorsally
Becomes a mucosa-lined vestibule	Becomes the anterior vaginal wall

- **TUM results:** popular, with excellent cosmesis and **no continence problems reported**, but **no long-term results.**
- **Concerns:** proximal circumferential dissection risks **sphincter or nerve injury**, and hence **stress incontinence**, and vaginal foreshortening.
- **Partial urogenital mobilisation (PUM, Rink)** uses the same circumferential dissection but **stops at the pubourethral ligament** (Fig. 60.25). It keeps the sinus for cosmesis and avoids retropubic and suprapubic dissection; it **suits the majority**, with conversion to TUM for a very high vagina.

TOTAL VERSUS PARTIAL UROGENITAL MOBILISATION

	TUM (Peña)	PUM (Rink)
Extent	Circumferential, above the pubis , with retropubic and suprapubic dissection	Circumferential, stops at the pubourethral ligament
Indication	Higher confluence; cloaca	Majority of urogenital sinus repairs
Theoretical risk	Sphincter or nerve injury, stress incontinence, vaginal foreshortening	Lower, since the aggressive dissection is avoided
Escalation	Pull-through if still high	Convert to TUM

KEY POINTS: UROGENITAL MOBILISATION

- TUM applies to both urogenital sinus and cloacal anomalies. The entire sinus is dissected circumferentially to above the pubis, so the confluence moves inferiorly to the perineum.
- With TUM or PUM, the mobilised sinus can form a mucosa-lined vestibule or an anterior or posterior vaginal wall, and should not be discarded.
- PUM mobilises the sinus circumferentially to the pubourethral ligament but not above it.

Concept — Mobilisation **brings the confluence to the surgeon** instead of taking the vagina off the urethra. It trades the hazardous anterior dissection for circumferential dissection near the sphincter, which is why PUM deliberately stops at the pubourethral ligament.

Results of Urogenital Sinus and Disorders of Sex Development Surgery**Limitations of the evidence**

- **The evidence base is weak:** limited, nearly always retrospective, small single-institution series, often with outdated or unreported techniques. One review judged the data too poor to interpret meaningfully.
- **Endpoints are inadequate.** Success is usually an open vagina with a feminised appearance; it should include **erotic sensation, lubrication, sexual satisfaction, orgasm and painless intercourse**, which are only beginning to be studied.
- The psychological effects of early surgery without consent, versus raising a child with atypia, are largely unexplored.
- **Cosmetic and early functional results are uniformly favourable; long-term outcomes are the concern.** Studies rarely match procedure to severity, and none report endocrine control.

Vaginoplasty outcomes**VAGINOPLASTY OUTCOME SERIES, GROUPED BY FINDING**

Finding	Bottom line
Revision after infant or prepubertal vaginoplasty	Common across series, and universal in some. Causes were failure to exteriorise the vagina and scarring
Azziz et al. (1986) , widely quoted	Satisfactory coitus in 62%: 46% of salt wasters versus 87% of non-salt-losers . Worse if operated before 1 year; many secondary procedures. Almost all had a cut-back , which leaves the narrow distal vagina
Secondary surgery near puberty	Usually successful; poorer results when the first operation was under 1 year
High confluence treated by pull-through	Most adults report satisfactory coitus; stenosis uncommon
Stenosis rates	From a minority to over a third, highest where cut-backs predominated ; dyspareunia common even without stenosis; dilation widely used

Finding	Bottom line
Early single-stage versus late	Far less stenosis after early single-stage repair than after late vaginoplasty; no difference before versus after 2 years. Planned one-stage beats planned multistage ; single-stage repair at 3–6 months advocated
Predictors	Confluence level and number of procedures predict outcome better than timing

- The pattern: **the worst results come from cut-back vaginoplasty** and failure to reach normal-calibre vagina, not from age at surgery. The trap is to quote Azziz against modern flap or pull-through repairs.
- **The distal vagina is always narrow**; current procedures must reach and exteriorise the normal-calibre vagina.
- Postpubertal surgery is tempting, but **no data show better outcomes**.
- **Early single-stage vaginoplasty often needs a secondary procedure after puberty**, as expected: the orifice stays the size it was made, since nothing flows through it in childhood. Secondary refinements are easy.

Clitoroplasty outcomes

- Technique has improved greatly on the basis of neuroanatomy, but there are **no long-term results for current techniques**.
- **Nerve conduction: pudendal evoked potentials** show modern clitoroplasty preserves conduction in the dorsal bundle, and sensation was intact in early series.
- **Conduction is not sensation**: absent sensation and orgasm can coexist with normal evoked potentials. Genitosensory testing has shown **markedly altered hot, cold and vibratory sensation**, while a small recent series found average sensation. **Sensation alone may poorly reflect clitoral normality**.
- **Cosmesis**: unsatisfactory in nearly half in one series; normal in a majority in another, with the rest too large, too small or absent. **No devascularisation** has been seen after modern clitoroplasty.
- **Satisfaction**:
 - Many operated women had multiple clitoral operations, and a fifth were dissatisfied, citing reduced sensitivity; yet **half of the non-operated thought the clitoris too large**.
 - **Orgasm did not differ** between operated and non-operated women.
 - Sensitivity is affected in nearly all after clitoroplasty, but **partial resection sparing more nerve function does no worse than no surgery**.
- **Physician–patient discrepancy**: in a European multicentre study, physicians rated results positively almost universally, while a minority of patients were (very) unsatisfied. Appearance and function correlated poorly. For clitoral size, about half were highly satisfied, a third indifferent, a minority dissatisfied.
- Whether initial virilisation affects sensitivity is unknown.

Urogenital mobilisation outcomes

- Results are **very early**. TUM and PUM are technically easier with superior cosmesis, but **functional superiority is unproven**. Stress incontinence or sphincteric denervation remains possible, although most report continence unaltered. **Use with caution** until long-term data exist.

Sexual function, mental health and lower urinary tract symptoms

- **Quality of life** and health-related quality of life are normal in most, with mental health equal to or better than population norms. A minority score lower because of distressing memories of **too-late surgery**, of the treatment itself, or poor sexual function. Relationships begin later than in the general population, and **patients prefer early surgery**.

- **Sexual satisfaction** is as good as in controls **except in salt-wasting CAH**, the most severely affected. Cosmetic score does not separate satisfied from dissatisfied women and does not influence sexual orientation. **Physicians consistently rate genital appearance better than patients do.**
- Feminising surgery **neither improves nor impairs psychosexual outcome**, especially with severe virilisation.
- **Homosexuality and bisexuality are more prevalent in CAH**, and more so in salt-wasting forms. Satisfaction with sex life is spread roughly evenly between satisfied, indifferent and dissatisfied.
- **Lower urinary tract symptoms** were a concern in CAH, but they do not differ from controls; occasional symptoms are common in both.

Timing of vaginoplasty revisited

- The few long-term studies suggest **delayed vaginoplasty may be preferable**, yet most children with CAH still have infant vaginoplasty, and most delay only the **small (under 3 cm), very high vagina**. Advocates of early surgery cite outdated techniques in those studies, the advantages of single-stage repair and easy secondary introitoplasty.

THE TWO-PROCEDURE DILEMMA

Approach 1	Approach 2
Simple infant procedure (clitorolabioplasty)	Extensive infant single-stage clitorolabiovaginoplasty
+ extensive postpubertal vaginoplasty	+ simple postpubertal introitoplasty

- Both approaches usually mean **two operations**, and **a moratorium would ensure the question is never answered.**
- **When to tell the child:** research is virtually non-existent.
- **CAH patients' own views:** opinion was divided on whether consented vaginoplasty in adolescence or adulthood is better than surgery before 6 months, with a plurality disagreeing; the most often preferred timing was **infancy**.
- **Golombok–Rust Inventory of Sexual Satisfaction (GRISS)** scores 1–4 normal and 5–9 increasing dysfunction. CAH patients scored higher than controls, but the proportion with individual dysfunction was similar; some had obsolete operations, including clitorectomy.
- **Elmasri et al. (2018) meta-analysis** for the Pediatric Endocrine Society guideline:
 - observational studies only; mean age at surgery 2.7 years; mean follow-up about 10 years;
 - **female gender identity in about 89%;**
 - Female Sexual Function Index about 25 of 36, indicating residual difficulty;
 - most sexually active and satisfied, but **only 1 in 2 had comfortable intercourse;**
 - **vaginal stenosis in 27%;** fistula, incontinence and urinary tract infection much less common;
 - **it does not compare early with late surgery.**
- **The most important factor affecting results is a surgeon or team with a special interest** in feminising genitoplasty. This supports centres of excellence; the CARES Foundation designates Comprehensive CAH Centers in the US.

KEY POINTS: UROGENITAL SINUS ANOMALIES — OUTCOMES OF RECONSTRUCTION

- Neonatal vaginoplasty usually needs a secondary procedure for stenosis after puberty (25–100%).
- Successful coitus is often reported; future studies should record sexual satisfaction, lubrication, erotic sensitivity and painless coitus.

- Current clitoroplasty seems unlikely to injure sensation, but sensitivity may not adequately reflect clitoral function, and some studies report reduced sensation with newer techniques.

Concept — The outcome literature cannot settle timing, because it measures **abandoned operations (cut-back)** against surrogate endpoints (**patency, appearance**). What it shows consistently is that **confluence level, number of procedures and surgeon expertise outweigh age at surgery**.

SURGICAL RECONSTRUCTION FOR CLOACAL MALFORMATIONS

Initial Management, Timing, and Principles

- **Stabilise first.** The infant is often ill, with abdominal distension that may compromise respiration.
- **Historical approach:** decompression, then rectal pull-through, with genitourinary reconstruction later or never; vaginal surgery was sometimes deferred to puberty for the benefit of oestrogen.
- **Majority view: piecemeal repair is a disservice and absolutely contraindicated.** Rectum, vagina and urethra should be repaired in a **single stage**. This gives the best exposure for rectovaginal separation, untouched tissue, and no need to remobilise the rectum.
- **Hendren's rule: rectal pull-through should never be a separate procedure.** Many of his referrals had had a previous pull-through, and definitive repair then needed repeat rectal mobilisation. Complete neonatal repair was also proposed, then doubted by its proponent.
- **Levitt and Peña's three neonatal pitfalls:** failure to recognise and manage **hydrocolpos; colostomy or vesicostomy problems; clinical misdiagnosis**.

SINGLE-STAGE VERSUS STAGED CLOACAL REPAIR

	Single-stage (majority)	Staged (dissenting groups)
Infant surgery	Rectal, vaginal and urethral repair together	Anorectal only , open or by combined endoscopic and laparoscopic pull-through
Urogenital repair	Same operation	Peripubertal ; Mitrofanoff if CIC needed
Rationale	Exposure, untouched tissue, no rectal remobilisation	Avoids early urethrovaginal complications that threaten the urinary and genital tracts and renal function; avoids unoestrogenised tissue; the patient shares the decision
Evidence claimed	—	Peripubertal vaginal reconstruction did not worsen voiding or continence, and fewer children needed CIC
Exclusion	—	Hydrocolpos excludes the laparoscopic staged approach

- **Four basic steps of management:**

Decompress the gastrointestinal tract → decompress the genitourinary tract → correct nephron-destructive or potentially lethal urinary anomalies → definitive repair of the cloaca.

Concept — The four-step sequence reflects one priority: **protect the kidneys and the child before reconstructing anatomy**. Definitive repair is never the first operation.

Decompression of the Gastrointestinal Tract

- A colostomy is agreed; its site is debated.

COLOSTOMY OPTIONS IN CLOACA

Colostomy	Proponent	Advantage	Drawback
Low loop	Hendren (initially)	—	Too low: bowel hard to obtain for vaginoplasty, and it interferes with rectal pull-through

Colostomy	Proponent	Advantage	Drawback
Right transverse, divided	Hendren (later)	Left colic blood supply intact; bowel available for vaginal replacement	Proximal site allows urine absorption
Descending, divided, just beyond the retroperitoneal attachment	Levitt and Peña	Less mucosa for urine resorption; enough distal colon for pull-through; less prolapse	Leave a long sigmoid, or the pull-through may need abdominal exploration
Proximal	Authors	Preserves distal colon	Hyperchloraemic acidosis from urine absorbed through colonic mucosa
None: balloon catheter drainage with irrigation three times daily	—	Allowed single-stage repair	—

- Colostomy siting trades **distal colon length** (pull-through, vaginal replacement) against **urine resorption and acidosis** and **prolapse**.
- **Malrotation** is recognised and corrected at colostomy; the **appendix is preserved** for a later appendicocaecostomy.
- **Endoscopy at colostomy** defines the anatomy and helps decompress the urinary tract.
 - The scope tends to **enter the vagina(s)**. The bladder can be extremely hard to reach, because the distended vagina compresses it against the anterior abdominal wall and obstructs the bladder neck.
 - **Draining and irrigating the vagina** relieves distension and may reveal the bladder.
 - The rectal fistula is hard to find: usually within the septum of duplicated vaginas, but it may be almost anywhere. Irrigating through the fistula and the **distal colostomy limb** clears inspissated mucus and meconium.
 - Leave **small feeding tubes in all channels** for fluoroscopic contrast studies in the radiology suite.

Decompression of the Genitourinary Tract

- After decompression, **voiding may continue into the vagina(s) and/or rectum**. This causes rapid distension, poor urinary drainage, persistent hydronephrosis, urinary tract infection (UTI) and **hyperchloraemic acidosis**.

CIC first-line → cut-back (opening the cloaca) if a long urethra-like cloaca defeats catheterisation (Fig. 60.26) → **vesicostomy or vaginostomy** only if CIC fails.

- **A catheter that enters the vagina rather than the bladder still decompresses** the genitourinary tract.
- **Unrecognised or mismanaged hydrocolpos** (most cases in one large series) causes ureteric obstruction, UTI, acidosis and failure to thrive, all reversed by adequate vaginal drainage.

GENITOURINARY DECOMPRESSION OPTIONS

Option	Position	Caveat
CIC	Authors' first line; usually keeps the vagina decompressed; equivalent to vesicostomy or vaginostomy in protecting the urinary tract and preventing hydrocolpos	May be difficult in a long urethra-like cloaca
Cut-back of the cloaca	Aids catheterisation	—
Vaginostomy or tube vaginostomy	Favoured by Levitt and Peña	No evidence it beats CIC. Requires an abdominal procedure and may tether the vagina to the abdominal wall , hampering later pull-through or TUM (a concern disputed by some)

Option	Position	Caveat
Vesicostomy	Preferred by some to avoid vaginal tethering	Only if CIC fails

Repair of Obstructive Urinary and Other Pathology

- This is the **third stage**. Structural renal abnormalities, some obstructive, occur in **60–83%** of persistent cloaca. With a **common channel over 3 cm**, urological defects are present in nearly all.
- **Do not defer obstructive lesions while awaiting definitive repair**; treat them promptly. Chronic kidney disease (CKD) occurs in **75%** and end-stage renal disease (ESRD) in **17%** of persistent cloaca.
- **Bladder function: urodynamics are abnormal in 90%** before and after repair. Patterns vary: overactivity, atonia, detrusor–sphincter dyssynergia, elevated leak point pressure.
- **Vesicoureteral reflux is common but managed medically** if UTI is prevented. Other systems (cardiac, spinal cord) are fully evaluated, and some are repaired in this phase.

Definitive Repair of Cloacal Malformations

- **Usually at 6–12 months**, in a well-nourished, thriving child; some operate at **1 month** in healthy infants.
- **Never before** the urinary tract has been evaluated, hydrocolpos excluded, and associated anomalies defined. A separate prior endoscopy helps.
- **Bowel preparation**: irrigate the distal colon through the mucous fistula and distal limb, plus a standard GoLYTELY preparation.

Operative Technique: Cloaca

- Some advise rectal pull-through only in infancy with later urogenital repair, but **most repair both together**; that technique is described here.
- **Prerequisites**: common-channel length, **urethral length**, vaginal anatomy and site of the rectal fistula.

WOOD ALGORITHM FOR THE UROGENITAL COMPONENT (ENDORSED BY THE AUTHORS)

Common channel	Urethral length	Procedure
Under 3 cm	and over 1.5 cm	TUM
Over 3 cm	or under 1.5 cm	Avoid TUM; urogenital separation, with the urethra reconstructed or the common sinus used as urethra
Any		Transabdominal TUM never: risks ischaemia of the channel

- TUM on a short urethra (under 1.5 cm) causes **urinary leakage**. **Normal female urethral length** on voiding cystourethrogram averages **2.43 cm** (about 2.3–2.6 cm across 6–36 months). A urethra **longer than 1.5 cm keeps the bladder neck above the urogenital diaphragm**.
- An alternative for common channel under 3 cm is to **start with PUM** and convert to TUM only if the urethra does not reach, which may benefit bladder function.

Preparation and posterior sagittal approach

Endoscopy → **Foley in the bladder**; **different-coloured Fogarty catheters in the vagina(s) and rectum** → circumferential preparation from nipples to toes, legs wrapped (allows prone and supine) → **prone jackknife**, pelvis on rolls → **electrical stimulator** locates maximal contraction, marked for the anus; perineal body and vaginal sites marked (Fig. 60.27) → an optional sound in the common channel (the catheters usually suffice).

- **Posterior sagittal anorectovaginourethroplasty (PSARVUP, Peña)**: its key advances are **identification of the external sphincter and muscle complex** and **excellent exposure** for separating rectum and vagina(s) from the cloaca.

- **Where is the rectum?** A rectal lumen **below the pubococcygeal line** suits the posterior sagittal approach, and one **below S3** is reachable from the perineum. **Higher** needs laparotomy or laparoscopy.

Rectal separation

Midline incision from the **coccygeal tip to the posterior cloacal orifice** → sphincter structures marked with sutures → common cloaca opened posteriorly to the confluences (**less appropriate with TUM**, as it limits use of the sinus) → rectal fistula opened posteriorly, multiple silk stay sutures → rectum separated from the vagina(s) → circumferential rectal mobilisation, well above the sacrum if needed, with peritoneal separation from below → rarely, supine abdominal mobilisation of a very high rectum → tension-free reach to the perineal skin → rectum retracted off the genitourinary structures.

- **Rectum and vagina share a common wall:** it is **better to enter the vagina** than the rectum or urethra. An abdominal approach is **more likely with a common channel over 3 cm**.

Classic vaginal separation (common channel over 3 cm or urethra under 1.5 cm)

- This is **even more tedious and difficult than rectal separation**, and entering the urinary tract is an error. **Side-by-side duplication** is most common, and its septum is incised with electrocautery at the initial endoscopy.

Fogarty palpated → posterior vagina opened at the confluence → circumferential stay sutures → **narrow malleable brain retractor** in the opened posterior wall exposes the anterior-wall confluence → **sharp dissection, not cautery** → traction on the stays mobilises the vagina(s) → vagina "pulled through" to near the perineum.

- The vagina frequently **almost encircles the urethra**; failure to recognise this injures both. **The outer vaginal wall is white. Avoid devascularising the vagina.**
- **If the vagina still does not reach:** use skin flaps to the spatulated vagina. For an extremely high vagina or vaginal agenesis, use **bowel interposition** (abdominal).
- The vaginal "switch" (a vaginal flap for very dilated vaginas) is **no longer recommended**, because of ischaemia and stenosis.

Urethra, perineum and rectum

Common-channel openings closed in **two to three absorbable layers** to form the urethra → **adjacent fat flap** over the repair (fistula prevention is critical) → a large, vagina-like cloaca is opened entirely and tailored over the Foley → vagina secured to the perineum, rotated to avoid overlapping suture lines if injured → **Foley left for 2 weeks** → perineal body reconstructed → rectum pulled through, widened rectum **tailored** → **muscles meet behind the rectum** → rectum centred in the sphincter mass and sutured to skin.

- **Aftercare:** rectal dilation from **2–3 weeks**; colostomy closure at about **3 months** if healed.
- **Laparoscopic assistance** gives low fistula dissection, accurate rectal placement and an excellent pelvic view. Single-incision surgery is as effective as three-port, with better cosmesis.
- **After catheter removal:** monitor voiding closely, because **poor emptying from a neuropathic component occurs in almost one-third**. Some leave a suprapubic tube until normal voiding is documented. Start **CIC at any doubt** and continue until bladder dynamics are normal; omission risks UTI, hydronephrosis and renal loss.

TUM in cloaca

- Peña's TUM is **widely accepted** within the algorithm above: easier and quicker, with less blood loss. The whole sinus is mobilised through 360 degrees **after rectal separation**, as in urogenital sinus TUM.
- Separating a vagina that still falls short after TUM was once practised but is **no longer recommended because of ischaemia**, which again stresses accurate preoperative measurement.
- Continence to date is no different from standard repair; long-term data are awaited.

KEY POINTS: CLOACAL RECONSTRUCTION

- Four steps: (1) gastrointestinal decompression (colostomy); (2) urinary decompression (CIC, vesicostomy, vaginostomy); (3) correction of associated organ-system anomalies; (4) cloacal repair (PSARVUP).
- TUM is now commonly used after rectal separation, but only when the common channel is under 3 cm and the urethra over 1.5 cm.
- Results depend largely on whether the spinal cord is normal. CIC and a bowel programme are frequently needed for urinary and faecal continence.

Concept — Common-channel length and urethral length decide whether the urogenital complex can be **moved en bloc (TUM)** or must be **taken apart (separation)**. Mobilising a channel whose urethra is too short drags the bladder neck below the continence mechanism.

Results of Cloacal Surgery

- Repair is exceedingly challenging: the anatomy is complex and individual, and the surgeon must be patient, imaginative and gentle. Nonetheless, **most patients lead productive lives**; the pioneers were **Hendren and Peña**.
- **Endpoints**: classically urinary continence, faecal continence and sexual capability. Future reporting should add erotic sensitivity, fertility, painless intercourse, orgasm and quality of life.

Urinary outcome

- Early results were dismal, with most patients incontinent.
- **A high proportion have a neuropathic component**, far more often when the **common channel exceeds 3.5 cm**.
- **High outlet resistance without neurological abnormality**: most ARM patients had preoperative leak point pressure above 40 cm H₂O, many with a normal spinal MRI.
- Among spinal abnormalities, a **short spinal cord** has the greatest impact on continence.

TABLE 60.1 CONTINENCE RATES AFTER CLOACAL SURGERY

Outcome	Range across series	Peña et al., 2004 (landmark)	Note
Voiding spontaneously	0.22–0.62	0.54	Hendren: body text gives 59% voiding and 28% CIC plus diversion, against the table's 0.62 and 0.34. Different denominator : the text counts all patients, including those too early to assess
Dry with CIC ± continent reconstruction	0.34–0.58	0.46	

Data from Hendren, 1998; Peña et al., 2004; Warne et al., 2002b; Rink et al., 2005b. CIC, clean intermittent catheterisation.

- The table's message: **at best about half void spontaneously**; the rest are dry only with CIC and/or reconstruction.
- **Peña et al. (2004), landmark series of 339 patients**: spontaneous voiding depended on common-channel length. **CIC was needed in 28% with a channel under 3 cm versus 78% over 3 cm**.
- **Good prognostic signs for spontaneous voiding**: a good bladder neck, short common channel, normal sacrum and two normal kidneys. A common channel over 3 cm also predicts worse renal outcome.
- After TUM, spontaneous voiding occurs in **43–66%**.

- **Voiding pattern disagreement:**
 - **non-contractile bladder**, warranting CIC (authors agree; inadequate detrusor contraction and neurovesical dysfunction predominate in other series);
 - **hyperreflexia predominant** in another series, in which most had an abnormal sacrum.
- **Largest pooled systematic review: under half void spontaneously (22–54%), 42% need CIC and 22% have urinary diversion.**

Renal outcome

- **CKD 75%, ESRD 17%.** Hydronephrosis and reflux are frequent, and most CKD is stage I–II. Other series report renal insufficiency in half, and an elevated or borderline creatinine in nearly half.
- **Aggressive diagnosis and modern bladder management** (CIC, diversion, augmentation) **can preserve renal function even with CKD.**
- Identify and treat renal impairment and obstruction at presentation. **Lifelong yearly urological evaluation**, with prompt treatment of obstruction or UTI and assessment of bladder dynamics.

Faecal outcome

- **Faecal continence is directly related to neurological status.** Hendren's good prognostic signs: good perineal raphe, well-defined anal dimple, normal spine and MRI, and brisk muscle reflex.
- Most have voluntary bowel movements, but **few never soil**. The incontinent are kept clean by a **bowel management programme**. The **Malone antegrade continence enema (MACE)**, via **appendicocaecostomy**, is used when bowel management fails.
- Long-term series report total incontinence in about a third, constipation in half and weekly soiling in a third; a few choose faecal diversion. Continence improves, and constipation lessens, into adolescence and adulthood.
- **More than half have major long-term defects of urinary and anorectal function.**

Gynaecological and sexual outcome

- **Complications of vaginal and perineal reconstruction:** introital stenosis, retained vaginal septum, persistent rectovestibular fistula, persistent urogenital sinus, and **inadequate perineal body**, generally from an **anteriorly placed rectum** at the initial PSARVUP. About half of patients referred after repair elsewhere need redo surgery.
- **Late vaginal stenosis and obstructed menstruation: 36–41%.** The vagina is adequate for intercourse in under half, and revision vaginoplasty is often needed. Normal menstruation occurs in about a third long term, and over a third need treatment for obstructive haematometra. Large cystic uterine or adnexal collections needing surgery also occur without obstruction.
- **Most reconstructed patients appear capable of normal sexual life.** Pregnancy occurs spontaneously and with in vitro fertilisation, with an **increased risk of preterm labour; caesarean section is preferred.**
- **Gynaecological evaluation** is needed at birth and again approaching puberty; uterovaginal anomalies are found in 53–67%.
 Ultrasonography **6–9 months after breast development** (thelarche precedes menarche by 1–3 years) → repeat every 6–9 months → until obstruction is excluded and periods are normal.
- **Leave atretic internal genital structures** at cloacal repair, since they may develop after hormonal exposure.

- **TUM series:** occasional urethral stenosis (managed with Mitrofanoff), urethrovaginal fistula and vaginal stenosis. Most children over 4 years were continent, about half voiding spontaneously and half by CIC; most had faecal continence, some via MACE.

Concept — Long-term outcome in cloaca is set more by the **spine and common-channel length** than by the operation. The operation's task is to avoid adding harm; the lifelong task is **surveillance of kidneys, bladder and menstrual outflow**.

SUMMARY

- Urogenital sinus and cloacal anomalies can be **exceedingly complex**.
- In DSD, **prompt but careful evaluation by an expert multidisciplinary team is mandatory**, with family support and education, openness and respect for child and family, respect for cultural practices, and confidentiality.
- At any age, reconstruction demands great care of tissues, meticulous attention to detail and a **lifelong commitment by the surgeon**, supported by a dedicated team.
- **The severe end of the spectrum should be treated only in centres of great expertise and experience.**